⑦ $\sum \left(\cos\frac{1}{\sqrt{n}} - \frac{\sqrt{n^2-n}}{n}\right)^{\alpha}$

$\frac{\sqrt{n^2-n}}{n} = \sqrt{1-\frac{1}{n}} = \left(1-\frac{1}{n}\right)^{1/2} = 1 - \frac{1}{2n} + \binom{1/2}{2}\frac{1}{n^2} + o\left(\frac{1}{n^2}\right)$

$\cos\frac{1}{\sqrt{n}} = 1 - \frac{1}{2n} + \frac{1}{4!}\frac{1}{n^2} + + o\left(\frac{1}{n^2}\right)$

$\Rightarrow \cos\frac{1}{\sqrt{n}} - \frac{\sqrt{n^2-n}}{n} = \left(\frac{1}{4!}+\frac{1}{8}\right)\frac{1}{n^2} + o\left(\frac{1}{n^2}\right) \sim \frac{1}{n^2} \Rightarrow$

$\sum\left(\cos\frac{1}{\sqrt{n}} - \frac{\sqrt{n^2-n}}{n}\right)^{\alpha} \sim \sum \frac{1}{n^{2\alpha}} \Rightarrow$ сх. при $2\alpha > 1$, т.е. $\alpha > \frac{1}{2}$

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⑧ $\sum\left(\frac{1}{\sqrt{n}} - \sqrt{\ln\frac{n+1}{n}}\right)^{\alpha}$

$\ln\frac{n+1}{n} = \ln\left(1+\frac{1}{n}\right) = \frac{1}{n} - \frac{1}{2n^2} + o\left(\frac{1}{n^2}\right) = \frac{1}{n}\left(1-\frac{1}{2n}+o\left(\frac{1}{n}\right)\right)$

$\sqrt{\ln\frac{n+1}{n}} = \frac{1}{\sqrt{n}}\left(1-\frac{1}{2n}+o\left(\frac{1}{n}\right)\right)^{1/2} = \frac{1}{\sqrt{n}}\left(1-\frac{1}{2n}+o\left(\frac{1}{n}\right)\right) = \frac{1}{\sqrt{n}} - \frac{1}{2n^{3/2}} + o\left(\frac{1}{n^{3/2}}\right)$

$\Rightarrow \frac{1}{\sqrt{n}} - \sqrt{\ln\frac{n+1}{n}} = \frac{1}{2n^{3/2}} + o\left(\frac{1}{n^{3/2}}\right) \sim \frac{1}{n^{3/2}} \Rightarrow$

$\sum\left(\frac{1}{\sqrt{n}} - \sqrt{\ln\frac{n+1}{n}}\right)^{\alpha} \sim \sum\frac{1}{n^{3/2\alpha}} \Rightarrow$ сх. ако $\frac{3}{2}\alpha > 1$, т.е. $\alpha > 2/3$.

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⑨ $\sum\left\{\left(e^{1/n} - \sin\frac{1}{n}\right)^{n^{\alpha}} - 1\right\}$

$\left.\begin{aligned} e^{1/n} &= 1 + \frac{1}{n} + \frac{1}{2n^2} + \dots \\ \sin\frac{1}{n} &= \frac{1}{n} - \frac{1}{6n^3} + \dots \end{aligned}\right\} \Rightarrow e^{1/n} - \sin\frac{1}{n} = 1 + \frac{1}{2n^2} + o\left(\frac{1}{n^2}\right) \Rightarrow$

$\left(e^{1/n} - \sin\frac{1}{n}\right)^{n^{\alpha}} = \left(1+\frac{1}{2n^2}+o\left(\frac{1}{n^2}\right)\right)^{n^{\alpha}} = e^{n^{\alpha}\ln\left(1+\frac{1}{2n^2}+o\left(\frac{1}{n^2}\right)\right)} =$

$= e^{n^{\alpha}\left(\frac{1}{2n^2}+o\left(\frac{1}{n^2}\right)\right)} = e^{\frac{1}{2}n^{\alpha-2}+o(n^{\alpha-2})} = 1 + \frac{1}{2}n^{\alpha-2} + o(n^{\alpha-2}) \Rightarrow$

$\left(e^{1/n} - \sin\frac{1}{n}\right)^{n^{\alpha}} - 1 \sim n^{\alpha-2} \Rightarrow \sum\{\sim\} \sim \sum n^{\alpha-2} = \sum\frac{1}{n^{2-\alpha}} \Rightarrow$

сх. ако $2-\alpha > 1$, т.е. $\alpha < 1$.

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⑩ $\sum \frac{\ln(e^n + n^2)}{n^2\ln^2(n+1)}$

$\ln(e^n + n^2) \sim \ln e^n = n$, понеже e^n е бързата ф-я, т.е. $\frac{e^n}{n^2} \xrightarrow[n\to\infty]{} \infty$

$\Rightarrow \sum \sim \sum\frac{n}{n^2\ln^2 n} = \sum\frac{1}{n\ln^2 n}$ – сходящ

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⑪ $\sum\left(1 - \frac{1}{\sqrt[n]{\ln(n+1)}}\right)$

$\frac{1}{\sqrt[n]{\ln(n+1)}} = (\ln(n+1))^{-\frac{1}{n}} = e^{-\frac{1}{n}\ln\ln(n+1)} = 1 - \frac{\ln\ln(n+1)}{n} + o\left(\frac{\ln\ln n}{n}\right)$

$\Rightarrow 1 - \frac{1}{\sqrt[n]{\ln(n+1)}} \sim \frac{\ln\ln(n+1)}{n} \sim \frac{\ln\ln n}{n} \Rightarrow \sum \sim \sum\frac{\ln\ln n}{n}$ – разх.

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⑫ $\sum\frac{\ln(n^2-n+1)}{\left(\arcsin\left(e^{1/\sqrt{n}}-1\right)\right)^{\alpha}}$

$\ln(n^2-n+1) \sim \ln n^2 = 2\ln n \sim \ln n$

$e^{1/\sqrt{n}} = 1 + \frac{1}{\sqrt{n}} + o\left(\frac{1}{\sqrt{n}}\right) \Rightarrow e^{1/\sqrt{n}} - 1 = \frac{1}{\sqrt{n}} + o\left(\frac{1}{\sqrt{n}}\right) \Rightarrow$

$\arcsin\left(e^{1/\sqrt{n}}-1\right) = \arcsin\left(\frac{1}{\sqrt{n}}+o\left(\frac{1}{\sqrt{n}}\right)\right) \sim \frac{1}{\sqrt{n}} = n^{-1/2}$

$\sum \sim \sum\frac{\ln n}{n^{-\alpha/2}} \Rightarrow$ сх. ако $-\frac{\alpha}{2} > 1$, т.е. $\alpha < -2$.

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⑬ $\sum\frac{\operatorname{arctg}\left(e - \left(1+\frac{1}{n}\right)^n\right)}{\ln^{\alpha}(1+n^2)}$

$\left(1+\frac{1}{n}\right)^n = e^{n\ln(1+1/n)} = e^{n\left(1/n - \frac{1}{2n^2} + o(1/n^2)\right)} = e^{1-\frac{1}{2n}+o(1/n)} = e.e^{-\frac{1}{2n}+o\left(\frac{1}{n}\right)}$

$\Rightarrow e - \left(1+\frac{1}{n}\right)^n = e\left(1 - e^{-1/2n + o\left(\frac{1}{n}\right)}\right) = e\left(1-\left(1-\frac{1}{2n}+o\left(\frac{1}{n}\right)\right)\right) = \frac{e}{2n} + o\left(\frac{1}{n}\right)$

$\Rightarrow \operatorname{arctg}\left(e-\left(1+\frac{1}{n}\right)^n\right) = \operatorname{arctg}\left(\frac{e}{2n}+o\left(\frac{1}{n}\right)\right) \sim \frac{e}{2n} \sim \frac{1}{n}$

$\ln(1+n^2) \sim \ln n^2 = 2\ln n \sim \ln n \Rightarrow \ln^{\alpha}(1+n^2) \sim \ln^{\alpha} n$

$\Rightarrow \sum \sim \sum\frac{1/n}{\ln^{\alpha} n} = \sum\frac{1}{n\ln^{\alpha} n} \Rightarrow$ сх. ако $\alpha > 1$

Сравнете с $\sum\frac{1}{n^{\alpha}\ln^{\beta} n} \to$ ако $\alpha = 1 \to$ сх. при $\beta > 1$.

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⑭ $\sum\frac{n^{\alpha}}{(\ln n)^{n/2}}$

Прилагаме Коши за $\sqrt[n]{a_n}$.

$\sqrt[n]{a_n} = \frac{(\sqrt[n]{n})^{\alpha}}{\sqrt{\ln n}} \to 0$, понеже $\sqrt[n]{n} \xrightarrow[n\to\infty]{} 1 \Rightarrow$ сх. $\forall\alpha$.

Или $(\ln n)^{n/2} = e^{n/2\ln\ln n} > e^n$ за достатъчно големи „n“.

Тогава $\sum\frac{n^{\alpha}}{(\ln n)^{n/2}} < \sum\frac{1}{e^n n^{-\alpha}} \to$ сходящ

Сравнете с $\sum\frac{1}{a^n n^{\alpha}\ln^{\beta} n}$ – сходящ ако $a > 1$.